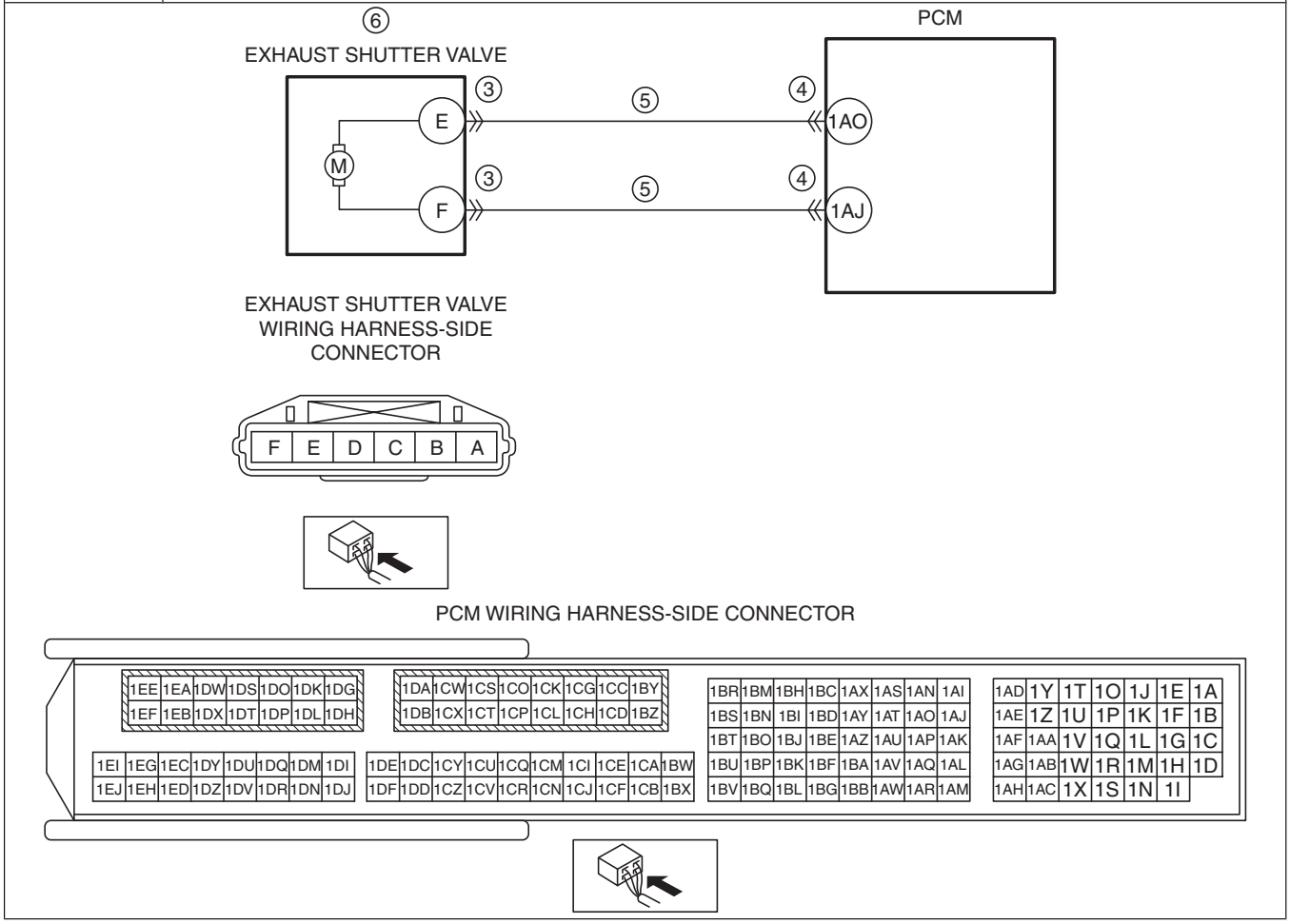
DTC P0476:00 [SKYACTIV-D 1.5]

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DTC P0476:00	Exhaust shutter valve control duty signal error
DETECTION CONDITION	• When the following condition is met, the exhaust shutter valve control duty value is **90%** or more for a continuos **2s.** **MONITORING CONDITIONS** — During engine running — The following DTCs are not detected: • PCM internal CPU: P0606:00 • Exhaust shutter valve DC motor control: P0475:00 • Exhaust shutter valve position sensor: P0408D:00, P048E:00 **Diagnostic support note** • This is a continuous monitor (EGR system). • The check engine light illuminates if the PCM detects the above malfunction condition during the first drive cycle. • FREEZE FRAME DATA/Snapshot data is available. • DTC is stored in the PCM memory.
FAIL-SAFE FUNCTION	• PCM restricts engine torque. • Inhibits the EGR control. • Inhibits the auto DPF regeneration control. • Inhibits the compulsory DPF regeneration control. • Inhibits engine-stop by operating the i-stop function.
POSSIBLE CAUSE	• Exhaust shutter valve connector or terminals malfunction • PCM connector or terminals malfunction • Open circuit in wiring harness between the following terminals: — Exhaust shutter valve terminal E—PCM terminal 1AO — Exhaust shutter valve terminal F—PCM terminal 1AJ • Exhaust shutter valve malfunction • Exhaust shutter valve position sensor malfunction • PCM malfunction



Diagnostic Procedure

STEP	INSPECTION		ACTION
1	**RECORD FREEZE FRAME DATA/SNAPSHOT DATA AND DIAGNOSTIC MONITORING TEST RESULTS TO UTILIZE WITH REPEATABILITY VERIFICATION** **Note** • Recording can be facilitated using the screen capture function of the PC. • Record the FREEZE FRAME DATA/snapshot data and DIAGNOSTIC MONITORING TEST RESULTS (A/F sensor related) on the repair order.	—	Go to the next step.
2	**VERIFY RELATED SERVICE INFORMATION AVAILABILITY** • Verify related Service Information availability. • Is any related Service Information available?	Yes	Perform repair or diagnosis according to the available Service Information. • If the vehicle is not repaired, go to the next step.
		No	Go to the next step.
3	**INSPECT EXHAUST SHUTTER VALVE CONNECTOR CONDITION** • Switch the ignition off. • Disconnect the exhaust shutter valve connector. • Inspect for poor connection (such as damaged/ pulled-out pins, corrosion). • Is there any malfunction?	Yes	Repair or replace the connector and/or terminals, then go to Step 7.
		No	Go to the next step.
4	**INSPECT PCM CONNECTOR CONDITION** • Disconnect the PCM connector. • Inspect for poor connection (such as damaged/ pulled-out pins, corrosion). • Is there any malfunction?	Yes	Repair or replace the connector and/or terminals, then go to Step 7.
		No	Go to the next step.
5	**INSPECT EXHAUST SHUTTER VALVE CIRCUIT FOR OPEN CIRCUIT** • Verify that the exhaust shutter valve and PCM connectors are disconnected. • Inspect for continuity between the following terminals (wiring harness-side): — Exhaust shutter valve terminal E—PCM terminal 1AO — Exhaust shutter valve terminal F—PCM terminal 1AJ • Is there continuity?	Yes	Go to the next step.
		No	Refer to the wiring diagram and verify whether or not there is a common connector between the following terminals: • Exhaust shutter valve terminal E—PCM terminal 1AO • Exhaust shutter valve terminal F—PCM terminal 1AJ **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for an open circuit. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has an open circuit. Go to Step 7.
6	**INSPECT EXHAUST SHUTTER VALVE POSITION SENSOR** • Reconnect all disconnected connectors. • Inspect the exhaust shutter valve position sensor. (See EXHAUST SHUTTER VALVE POSITION SENSOR INSPECTION [SKYACTIV-D 1.5].) • Is there any malfunction?	Yes	Inspect the exhaust shutter valve, then go to the next step . (See EXHAUST SHUTTER VALVE INSPECTION [SKYACTIV-D 1.5].)
		No	Go to the next step.
7	**VERIFY DTC TROUBLESHOOTING COMPLETED** • Always reconnect all disconnected connectors. • Clear the DTC from the PCM memory using the M-MDS. (See CLEARING DTC [SKYACTIV-D 1.5].) • Perform the KOEO or KOER self test. (See CLEARING DTC [SKYACTIV-D 1.5].) • Is the same DTC present?	Yes	Repeat the inspection from Step 1. • If the malfunction recurs, replace the PCM. (See PCM REMOVAL/INSTALLATION [SKYACTIV-D 1.5].) Go to the next step.
		No	Go to the next step.

STEP	INSPECTION		ACTION
8	**VERIFY AFTER REPAIR PROCEDURE** • Perform the “AFTER REPAIR PROCEDURE”. (See AFTER REPAIR PROCEDURE [SKYACTIV-D 1.5].) • Are any DTCs present?	Yes	Go to the applicable DTC inspection. (See DTC TABLE [SKYACTIV-D 1.5].)
		No	DTC troubleshooting completed.